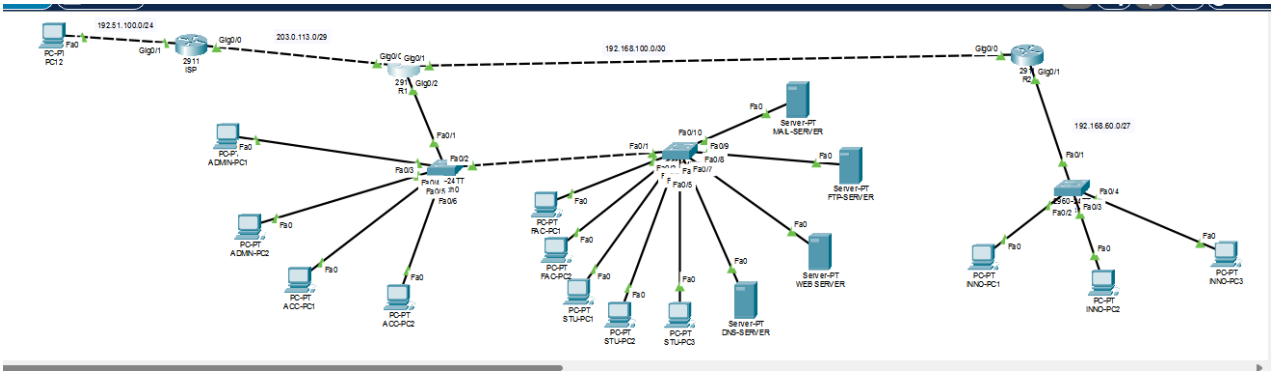
Complex Engineering Project – Computer Networking Lab (CSE342)

Southeast University | Summer 2026

1. Network Topology



Two buildings connected through Router R1 and R2, with an ISP router on the outside representing the internet (PC12).

2. IP Addressing Table

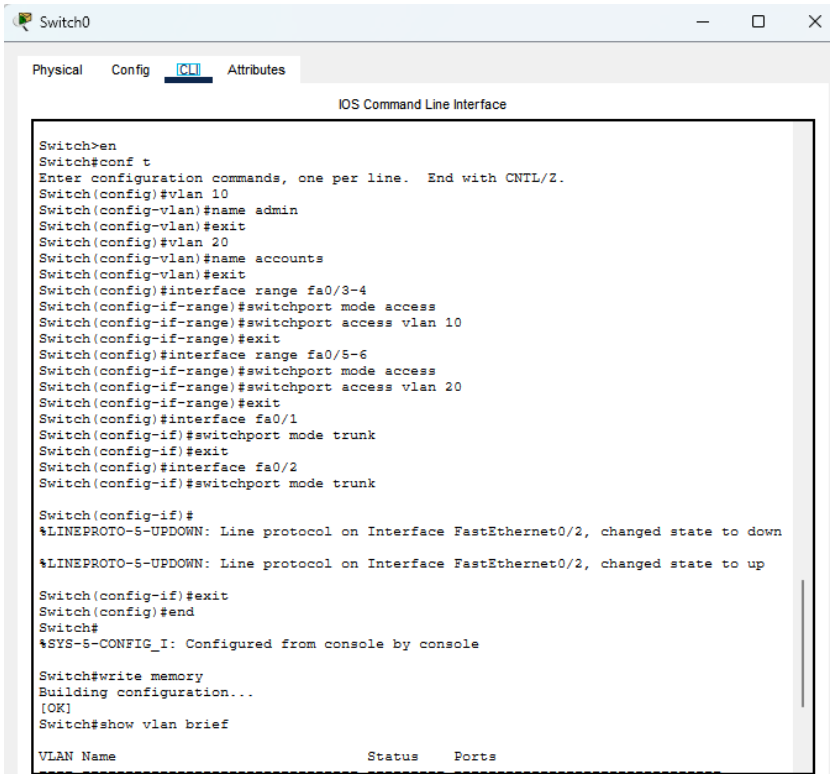
Private block 192.168.0.0/16 was divided into subnets for each unit using VLSM, based on the number of hosts.

Unit	VLAN	Network Address	Subnet Mask	Usable Host Range	Gateway
Admin	10	192.168.10.0	/27	192.168.10.1 - .30	192.168.10.1
Accounts	20	192.168.20.0	/28	192.168.20.1 - .14	192.168.20.1
Faculty	30	192.168.30.0	/27	192.168.30.1 - .30	192.168.30.1
Student Lab	40	192.168.40.0	/25	192.168.40.1 - .126	192.168.40.1
Server Room	50	192.168.50.0	/28	192.168.50.1 - .14	192.168.50.1
Innovation Lab	60	192.168.60.0	/27	192.168.60.1 - .30	192.168.60.1
R1-R2 Link	-	192.168.100.0	/30	192.168.100.1 - .2	-
R1-ISP Link	-	203.0.113.0	/29	203.0.113.1 - .6	-

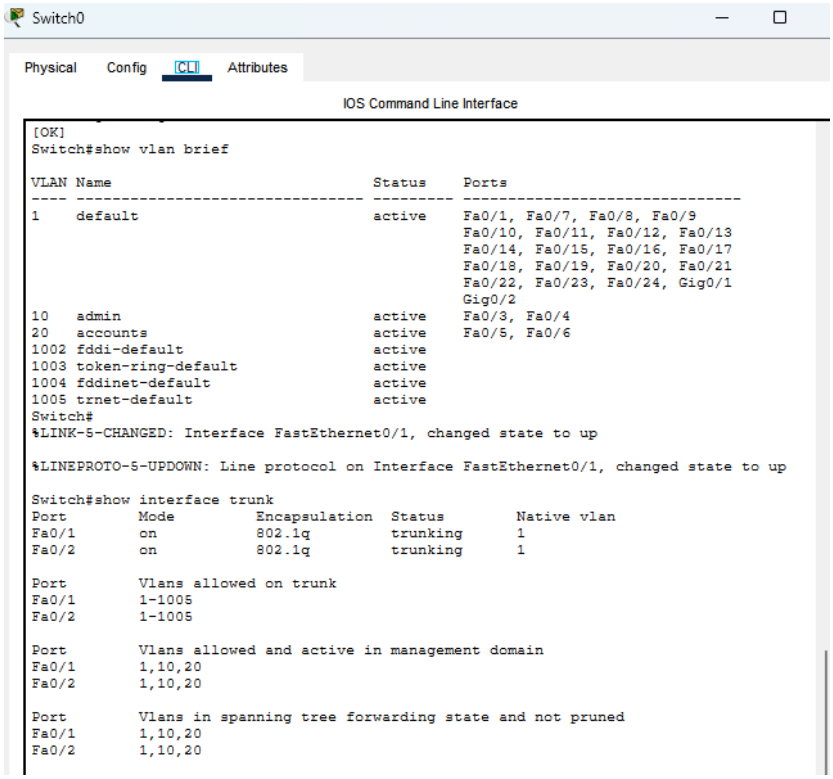
Admin, Faculty and Innovation Lab use /27 (30 hosts each). Accounts and Server Room use /28 (small units). Student Lab needs 100 hosts so /25 was used. Router links use /30. Note: Faculty subnet has no extra room for future hosts; could be widened to /26 later.

3. VLAN & Trunk Configuration

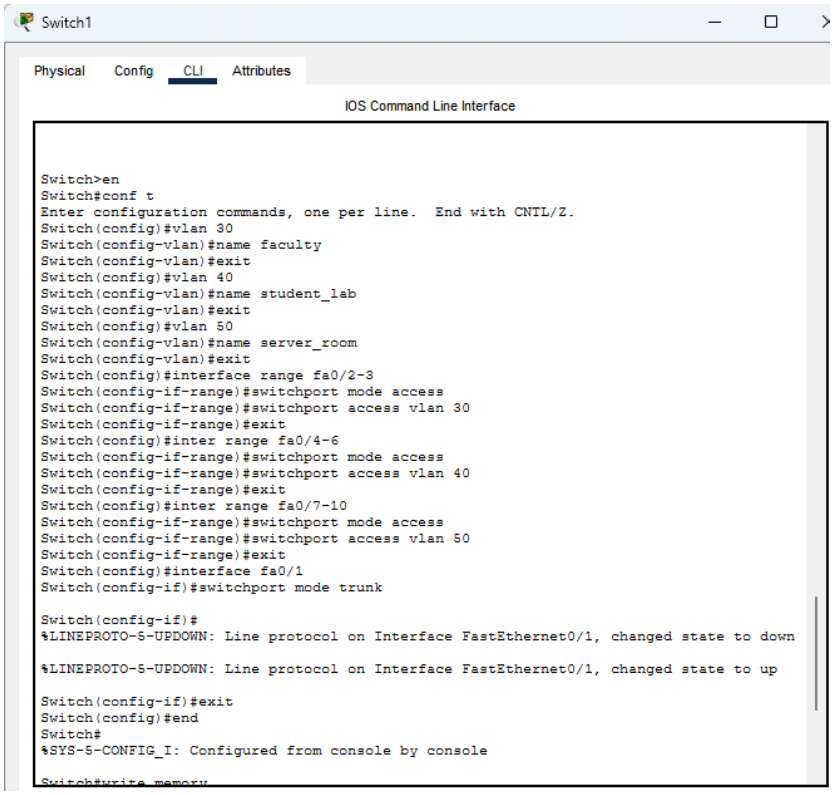
3.1 Switch0 (SW1) - Admin + Accounts: Configuration



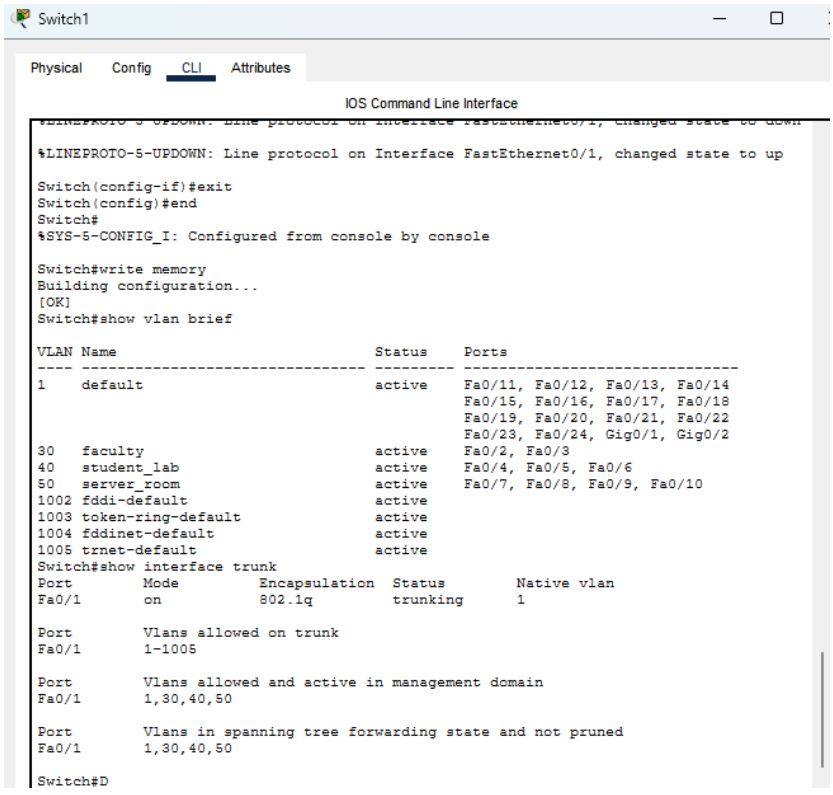
3.1 Verification - vlan brief and trunk status



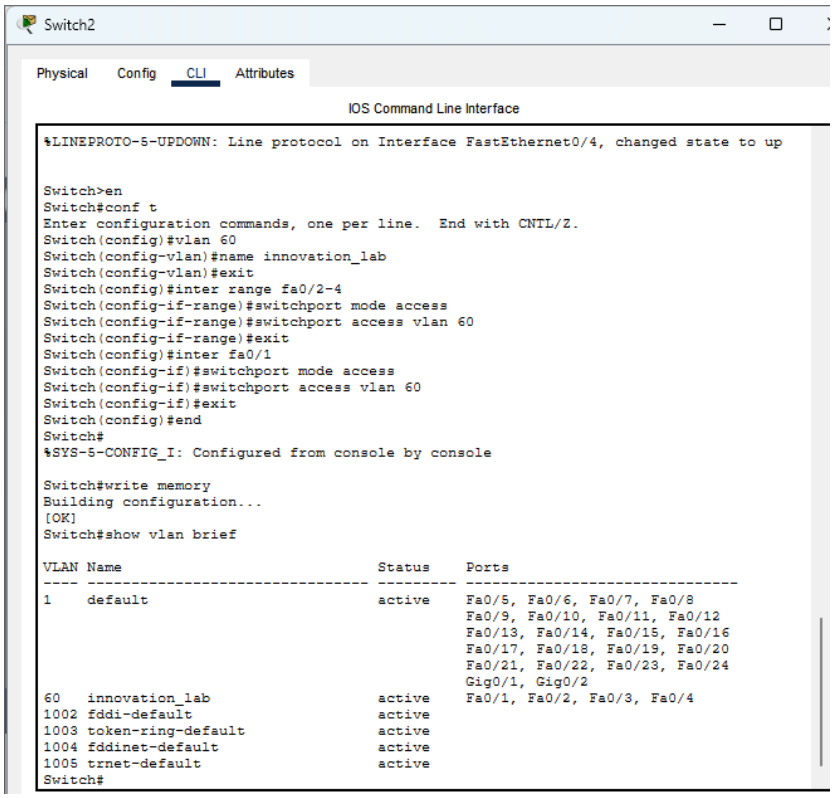
3.2 Switch1 (SW2) - Faculty, Student Lab, Server Room: Configuration



3.2 Verification - vlan brief and trunk status

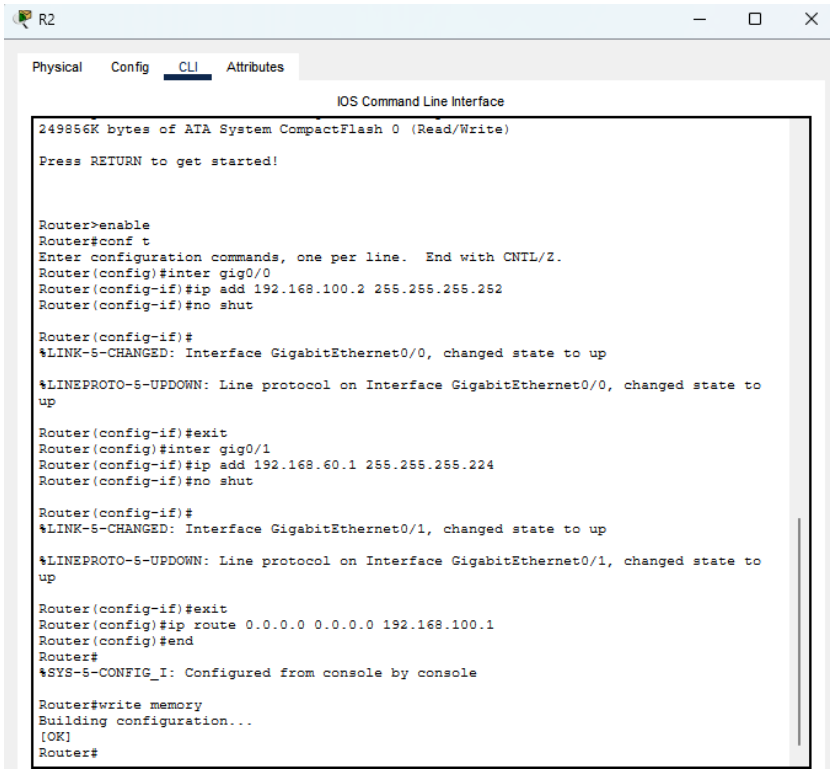


3.3 Switch2 (SW3) - Innovation Lab (config + verification)

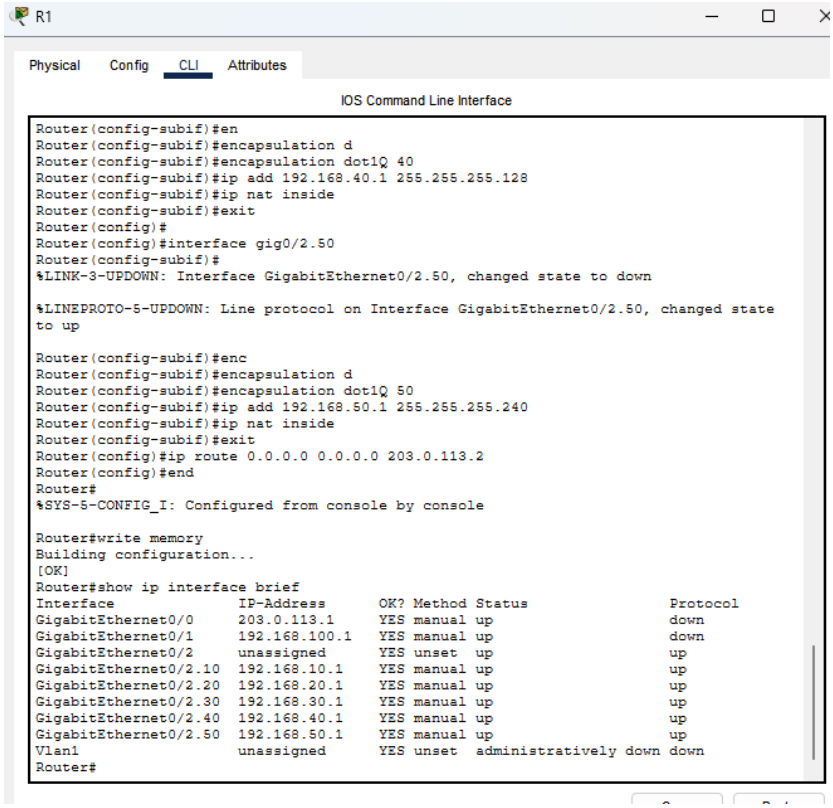


4. Router Configuration

4.1 R2 - Base Interfaces

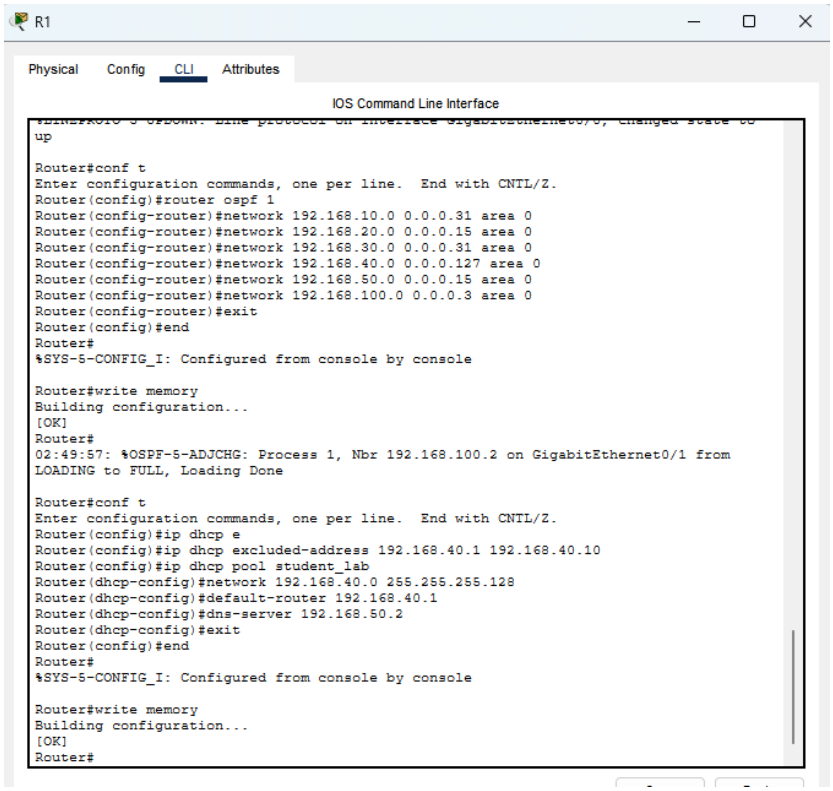


4.2 R1 - Subinterfaces (Router-on-a-Stick) + Verification

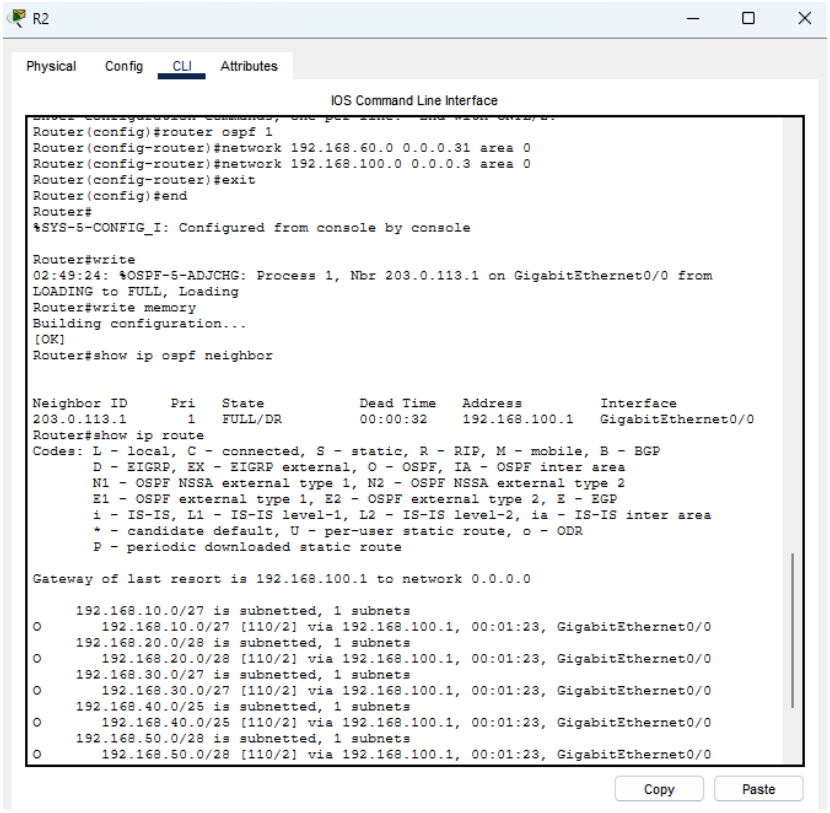


5. OSPF Routing and DHCP

5.1 R1 - OSPF and DHCP Pool Configuration



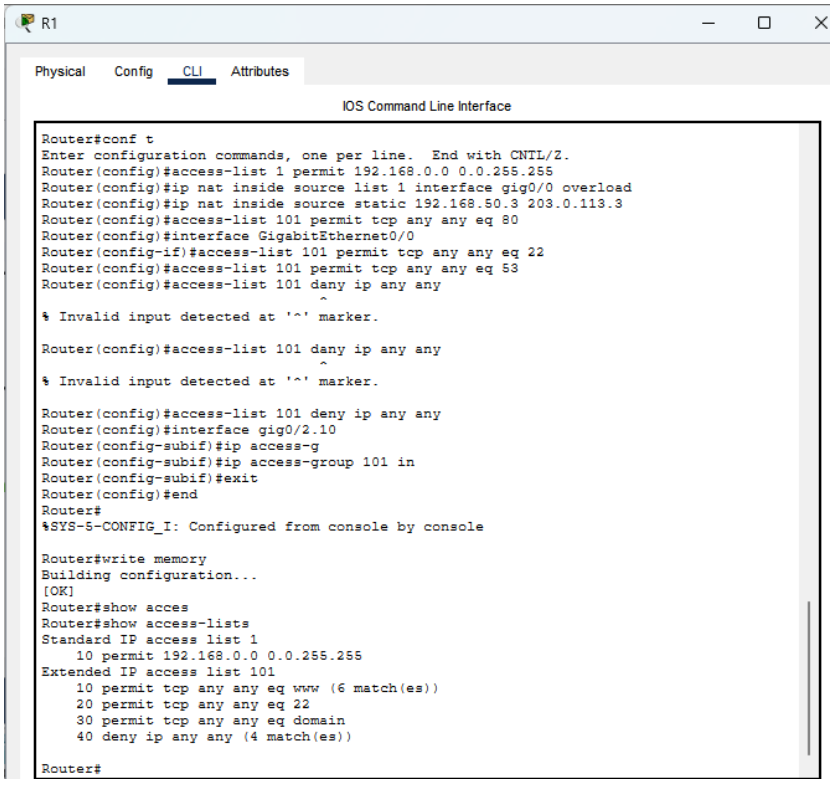
5.2 R2 - OSPF Configuration and Verification



OSPF neighbor with R1 is FULL/DR; all remote VLAN subnets appear as 'O' routes, confirming inter-building routing works.

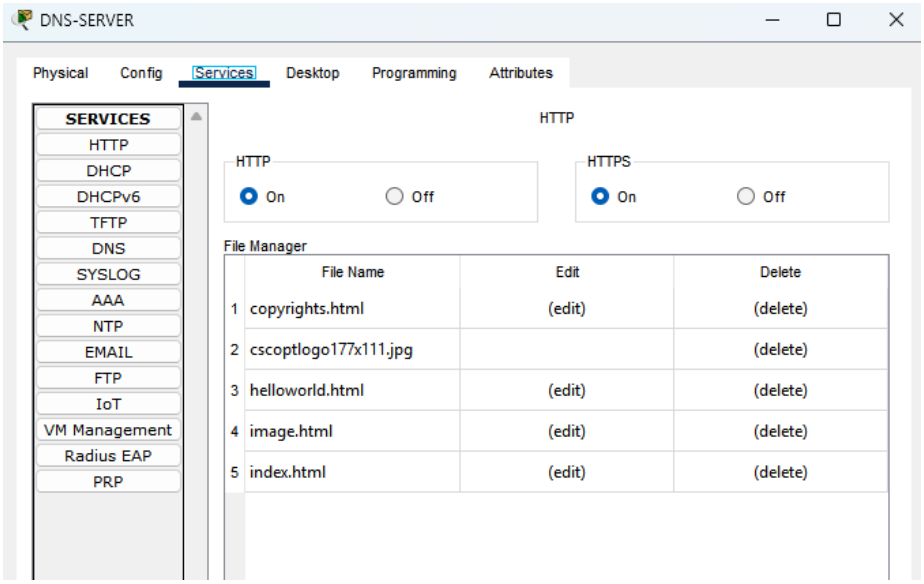
7. NAT / PAT and ACL Configuration

7.1 R1 - NAT (PAT + Static) and ACL Configuration

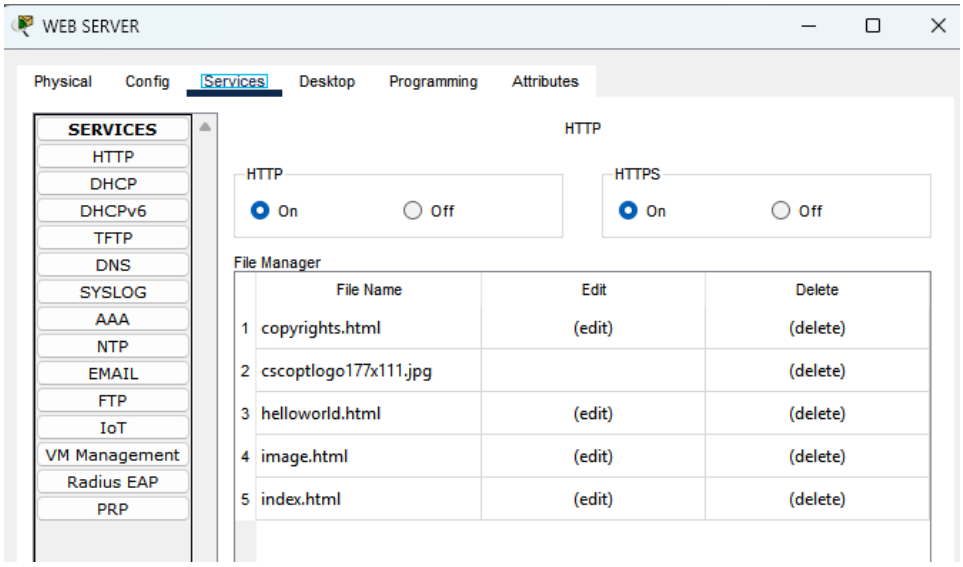


6. Network Services (Server Room)

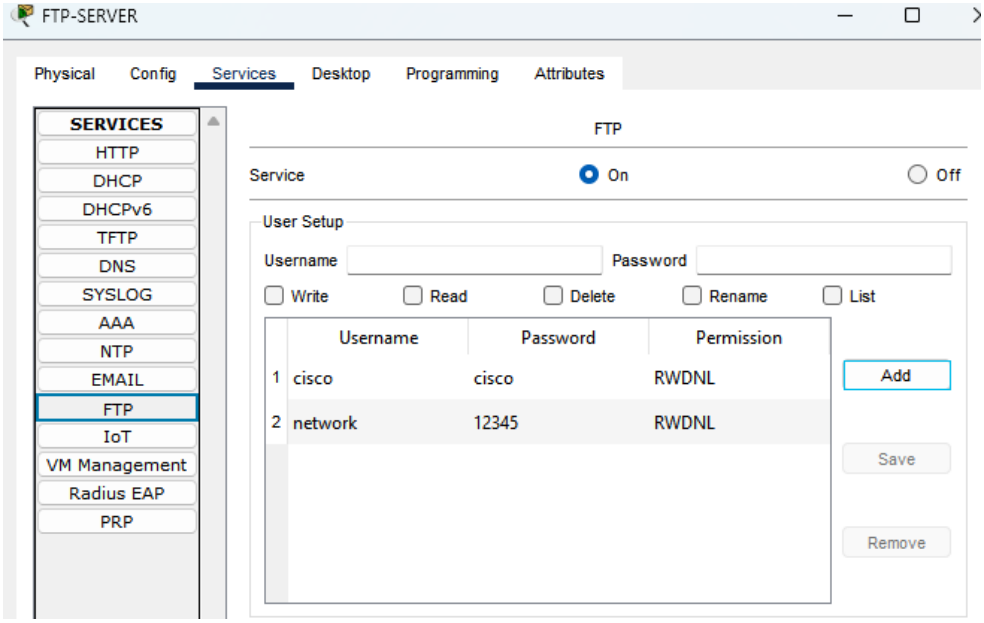
6.1 DNS Server - HTTP/DNS Service



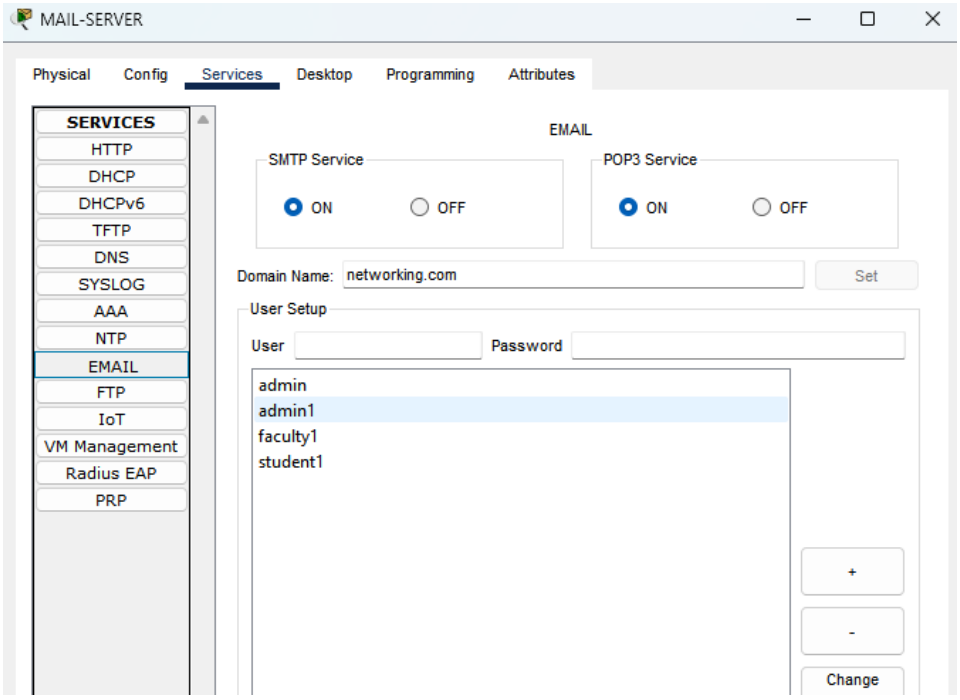
6.2 Web Server - HTTP Service



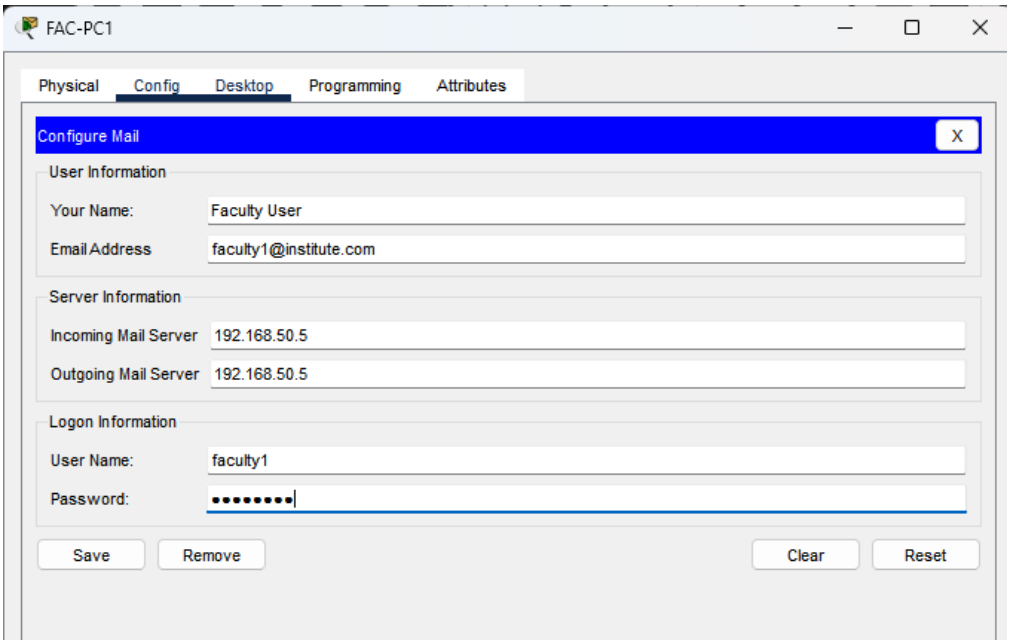
6.3 FTP Server



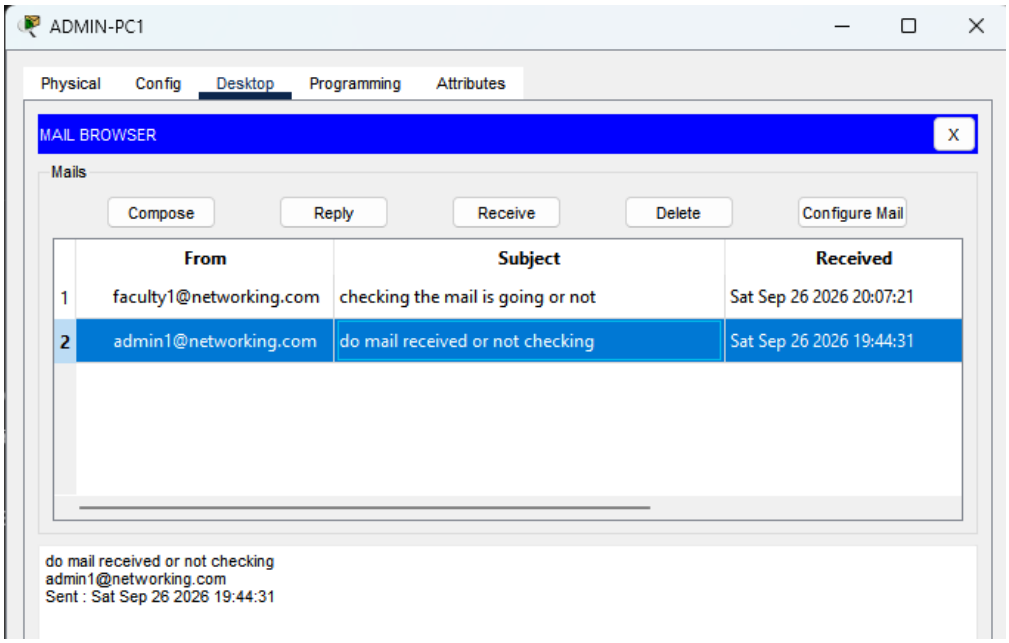
6.4 Mail Server - SMTP/POP3 Service



6.5 Mail Client Setup (Faculty PC)

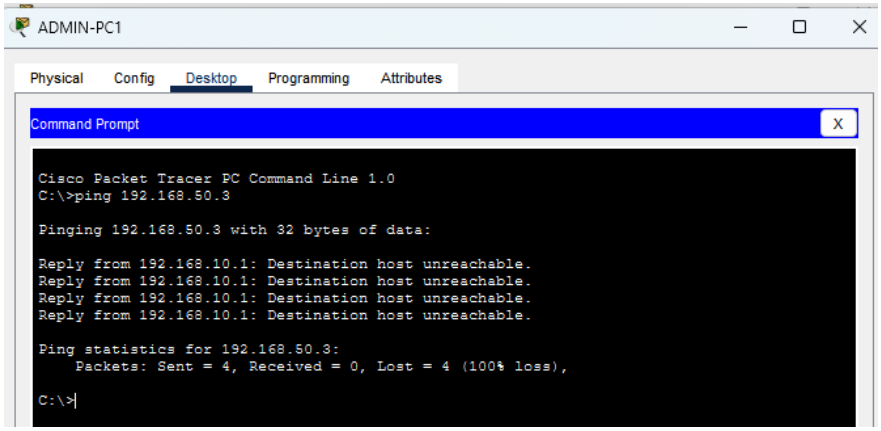


6.6 Mail Test - Inbox Showing Sent/Received Mail



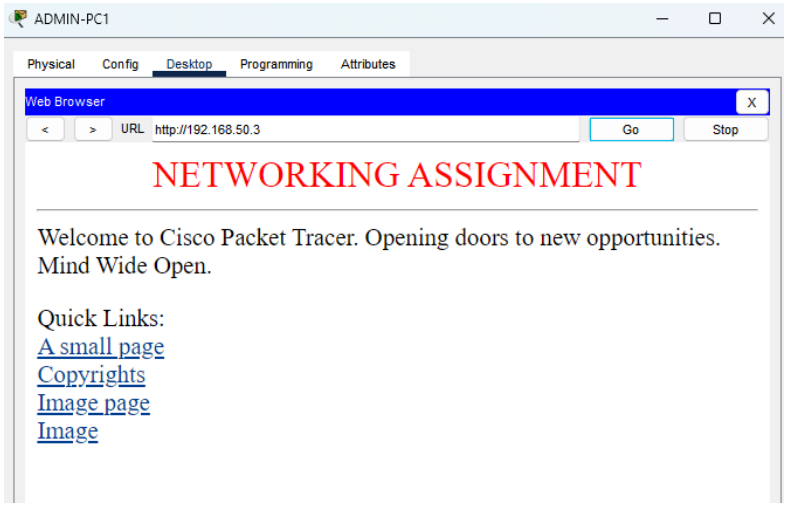
Test mail was sent between Admin and Faculty users and received successfully.

7.2 ACL Test - Ping Blocked on Admin VLAN



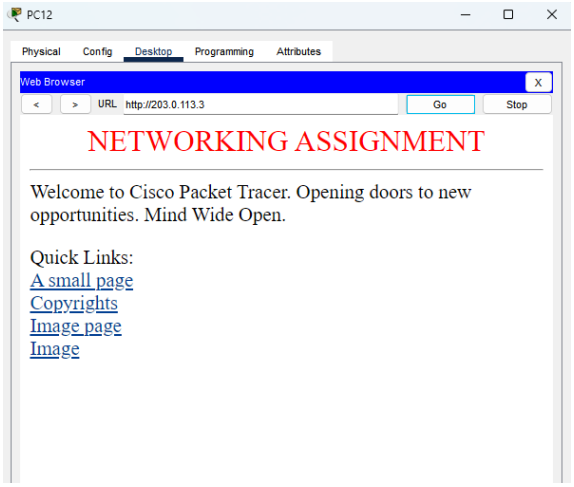
ADMIN-PC1 cannot ping the Web Server - ICMP is not permitted by ACL 101.

7.3 ACL Test - HTTP Allowed on Admin VLAN



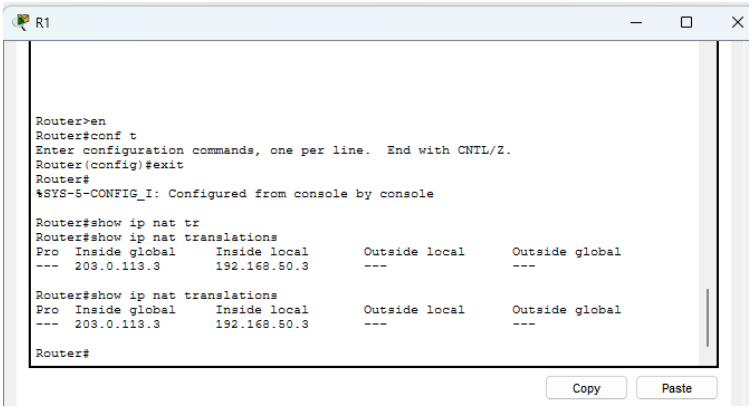
ADMIN-PC1 can still open the Web Server page over HTTP, allowed by ACL 101.

7.4 Static NAT - External Reachability Test (from PC12, outside network)



PC12 (outside the institute network, on the ISP side) opened http://203.0.113.3 and reached the Web Server page. This proves the web server is reachable from the simulated external network, as required by the scenario.

7.5 NAT Translation Table (R1)



The static entry (203.0.113.3 ↔ 192.168.50.3) confirms the Web Server's static NAT mapping is active on R1.

8. Troubleshooting Record

During configuration, a few issues came up. These were investigated and fixed as shown below.

Problem	Investigation	Root Cause	Correction	Verification
Router rejected a secondary IP on Gig0/0	Checked interface config on R1	Primary and secondary address were in the same subnet	Removed the extra secondary address line	'show ip interface brief' showed the interface up with a single correct IP
Wrong IP applied on Gig0/2.10 ('overlaps' error)	Compared configured IP with the addressing plan	VLAN 20's address was typed by mistake on the VLAN 10 sub-interface	Removed the wrong address and applied the correct VLAN 10 address	Interface came up with the correct 192.168.10.1 address
Student Lab PCs could not get an IP from DHCP	Checked VLAN on switches, trunk status and DHCP pool on R1	VLAN 40 was never created on SW1, so the trunk did not forward VLAN 40 traffic to R1	Created VLAN 30, 40 and 50 on SW1 (no port assignment needed)	Trunk showed VLAN 40 active; Student Lab PCs received IP addresses correctly

9. Suggested Improvement

Currently R1 is the only path between the two buildings and the ISP. Adding a second redundant link between R1 and R2, along with basic network monitoring, would make the network more reliable if a link or device fails in the future.